Security Analysis: Bauloo Styles

RAT and malware review of the Minecraft Fabric mod (version 1.6.0, MC 1.21.11) and comparison of the official JAR files with the published source code

Completed	20 October 2026, 00:53 CEST (UTC+2)
Type of review	**Independent review out of personal interest.** No connection to the creator of the mod (see section 10)
Scope	Repository **bauloo-styles-main** (source code), repository **bauloo-styles-cloud-main** (releases, packs, tools), two JAR files from the official website
Method	Static code analysis, bytecode analysis, decompilation (Vineflower 1.11.1), reproducible build from source and byte-for-byte comparison
Prepared by	Riyu(FPS) in cooperation with Claude Code (Opus 5.5)
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Result: No RAT, no stealer, no malware found.

The program code of the official JAR file **bauloo-styles-1.6.0-mc1.21.11-auto-update.jar** is **byte-identical** to what is built from the published source code: all 164 class files have identical SHA-256 checksums. The source code the author published as proof is therefore exactly the code that runs on your PC.

The second file **bauloo-styles-1.6.0-mc1.21.11.jar** differs in only 4 classes: the Spotify feature has been removed from it. It contains no additional code.

However, a few **privacy points** were found (telemetry including the Minecraft name, remote blocking, presence lookup). They are described in section 7 but are not malicious functions.

Independent review

Riyu(FPS) examined the mod together with Claude Code **out of personal interest and independently of the creator**. There is no collaboration, no commission and no other connection with the creator of the mod or their projects. The result applies only to the files reviewed at the time of review; there is no guarantee for future versions (disclaimer in section 10).

Contents

- 1. Files examined
- 2. Approach
- 3. Official JARs vs. repository (core result)
- 4. Difference between the two website variants
- 5. Code analysis: checking for RAT and stealer traits
- 6. Cloud repository, packs and build environment
- 7. Privacy notes (not malware)
- 8. Limitations of the analysis
- 9. Recommendations
- 10. Independence and disclaimer
- 11. Contact
- Appendix A: Checksums
- Appendix B: Tools and reproducibility

1. Files examined

File / folder	Content	Size
bauloo-styles-1.6.0-mc1.21.11-auto-update.jar	Official mod file from the website (variant with Spotify)	10,123,495 bytes
bauloo-styles-1.6.0-mc1.21.11.jar	Official mod file from the website (variant without Spotify)	10,114,133 bytes
bauloo-styles-main/	Source code of the mod: Java, resources, Gradle build	111 Java files, 21,038 lines; 571 images
bauloo-styles-cloud-main/release/	Ready-built JARs for 12 Minecraft versions, 2 variants each, plus version.json	24 JARs
bauloo-styles-cloud-main/packs/	Texture packs that the mod downloads on demand	301 ZIPs
bauloo-styles-cloud-main/previews/	Preview images of the packs	301 PNGs
bauloo-styles-cloud-main/tools/	The author's publishing script (stylescloud.py) and compiled .pyc	2 files

The two website JARs are byte-identical to the files in the cloud repository:

Website file	identical to	SHA-256
…-auto-update.jar	release/bauloo-styles.jar	`20684bceb237774ae1bc9b208216b057d3f93fbebdf9b4cfa72cdb5856d6c19f`
…-mc1.21.11.jar	release/bauloo-styles-no-auto-update.jar	`3056d0be0cb5e4daa3d3b7830be551036a0e4c3b4976e922fe1b531cafa23b7a`

2. Approach

The review consisted of several independent steps. None of the examined files were executed, with the exception of the Gradle build from source. Its build scripts were read beforehand; the build ran in an isolated copy with its own Gradle cache.

- **Source code review:** Search for typical malicious code patterns (launching processes, loading code at runtime, access to browser, Discord or launcher data, session tokens, keyboard/screen, encryption/obfuscation) and manual reading of every class with network or file access.
- **Bytecode analysis:** A custom script reads the constant pools of all classes in all 26 JARs and lists every method called and every string. No code is loaded or executed in the process.
- **Structural check of the JARs:** Foreign classes, embedded JARs, DLL/EXE files, manifest and Fabric entry points.
- **Decompilation:** Both website JARs with Vineflower 1.11.1 (checksum verified against Maven Central).
- **Reproducible build:** The source code was built with the included Gradle wrapper (Gradle 9.6.1, Fabric Loom 1.17.16, JDK 21.0.12). The self-built JAR was compared with the official JARs at the level of individual files (SHA-256) and at the level of decompiled code.
- **Contents:** All 301 packs against the checksums in index.json, file types, path tricks (Zip Slip), magic bytes and appended data in images; .pyc against .py; Gradle wrapper against the official checksums from gradle.org.

3. Official JARs vs. repository (core result)

A direct comparison of decompiled code with the original source code is only possible roughly: comments are missing, the compiler inlines constants (e.g. `MOD_ID` becomes `"bauloo_styles"`), and in the JAR, Minecraft methods carry intermediary names such as `method_25394` instead of `render`. Such a rough comparison showed only this expected noise.

That is why the source code was also built independently. If the built bytecode matches the official JAR, this proves that the JAR was produced from exactly this source code. Result:

Comparison	Result
Class files (.class), official auto-update JAR vs. own build	**164 of 164 byte-identical** (identical SHA-256)
Decompiled code, official auto-update JAR vs. own build	**no differences** in 111 Java files (21,892 lines)
Resources outside resourcepacks/	772 of 779 byte-identical; 7 differ, see below
Files only in the official JAR	1,170 files in resourcepacks/ (97 built-in texture packs)
Files only in own build	none

Explanation of the differences:

- `fabric.mod.json`: Only the version string differs (the official build script appends the MC version). Entry points, mixins and dependencies are the same.
- 5× `respawn_anchor_N.json` and `tornado.png.mcmeta`: identical in content (compared as JSON). The bytes differ only in line endings (Windows CRLF vs. Unix LF).
- `resourcepacks/`: According to the README, the 97 built-in packs are deliberately not included in the repository (rights of the pack creators). They were checked separately: only images, JSON and .mcmeta, plus three harmless Windows system files (desktop.ini, 2× thumbs.db). No classes, no executable files.

```
diff official-auto-update/fabric.mod.json  repo-build/fabric.mod.json
<   "version": "1.6.0+mc1.21.11",
>   "version": "1.6.0",
<   "minecraft": "1.21.11",
>   "minecraft": ">=1.21.11",
```

4. Difference between the two website variants

The file names suggest a difference in automatic updating. That is not the case: both variants contain exactly the same update code, and since version 1.6.0 it only checks whether a new version exists. It downloads nothing and replaces no files.

The actual difference concerns only 4 classes, all related to the Spotify feature:

Class	auto-update (= repository)	without suffix
SpotifySmtc	Reads title, artist, position and cover art of Spotify playback via JNA and the Windows media controls (combase/WinRT)	Empty placeholder, always returns null
SpotifyNowPlaying	Background thread; fallback reads window titles (User32.EnumWindows) and filters for processes named spotify.*	AVAILABLE = false, start() does nothing
SpotifySmtc$Info	Data class	same content (only compiler details differ)
BaulooStylesConfigScreen	Menu contains Spotify tile and toggle	Spotify tile and toggle removed

The variant without Spotify probably exists because antivirus scanners tend to flag window enumeration and native Windows calls as suspicious. The Spotify feature is **off** by default (`spotifyHud = false`). As long as it is off, no windows are read and no requests are sent to Deezer/iTunes.

5. Code analysis: checking for RAT and stealer traits

Typical traits of Minecraft mod malware (session stealers, Discord token grabbers, RATs) and the findings in the code. The check covered the source code and, in addition, the bytecode of all 26 JARs.

Trait	Finding
Launching programs (Runtime.exec, ProcessBuilder)	Not present
Loading code at runtime (ClassLoader, defineClass, URLClassLoader, script engines)	Not present
Reflection / Class.forName	Only for optional compatibility with the mods *Blur* (BlurConfig) and *Sodium Extra*
Minecraft session token (getAccessToken / getSessionId)	Not present. Only the **name** (method_1676) and **UUID** (method_44717) are read
Browser data, cookies, passwords (Login Data, Local State)	Not present
Discord tokens (leveldb), launcher accounts (launcher_accounts.json)	Not present
Crypto wallets	Not present
Keyloggers, screenshots (Robot, GlobalScreen)	Not present
Clipboard	Only in the menu: copying/pasting preset codes by the user
Obfuscation, Base64, encryption	Not present; nothing is obfuscated
Native libraries / DLLs in the JAR	None; JNA ships with Minecraft and is only used for Spotify
Embedded JARs (jar-in-jar)	None
Incoming connections / remote control	No remote access; local server only on 127.0.0.1 (see 5.3)
Self-update / downloading program code	No, only a version check with a notice in the menu

5.1 Network connections

Complete list of all destinations present in the bytecode:

Destination	Purpose	Data sent
raw.githubusercontent.com, api.github.com, cdn.jsdelivr.net (repo Bauloo0/bauloo-styles-cloud)	Style list, packs, preview images, version check, block list	No personal data
bauloo-styles-cloud.bauloo-styles-cloud-worker.workers.dev (the author's Cloudflare Worker)	Mirror for downloads, statistics, tab icons, presets, feedback, VIP features	Minecraft name, hash of the device ID, UUIDs (details in section 7)
textures.minecraft.net	Skin/cape textures	Texture hash
api.deezer.com, itunes.apple.com	Album covers for the Spotify display (only if enabled)	Artist and song title
baulooprac.xyz	Feedback page, opened in the browser	One-time token in the URL
127.0.0.1 (local)	Upload page for your own packs	Stays on the PC

5.2 File access

The mod only reads and writes in these folders:

- `.minecraft/config/bauloo-styles/`: cache, settings, your own uploads, capes
- `.minecraft/resourcepacks/`: downloaded and your own style packs (prefix bauloo_)
- `%USERPROFILE%/.bauloo-styles/`: random device ID (device-id.txt), optionally a publish key of the user

- Temporary files in the system temp folder during downloads

Directories are only traversed within these folders (Files.walk / Files.list). No other areas of the hard drive are read.

5.3 Local web server (upload page)

A small HTTP server in `StylesUploadServer` only starts when the user opens the upload page. It is properly secured:

- Binds only to the loopback address (`InetAddress.getLoopbackAddress()`); not reachable from the network.
- Random port and random 128-bit token (SecureRandom), compared in constant time.
- Checks the Host and Origin headers; protects against DNS rebinding and third-party websites.
- Shuts down after 30 minutes of inactivity.

5.4 Downloading and processing packs

- Every pack is checked against the size and SHA-256 from index.json, so a mirror server cannot slip in different files.
- Only entries under `assets/` (or `assets/minecraft/optifine/sky/` for skies) are taken from the ZIP; paths containing `..` or a leading `/` are discarded (protection against Zip Slip).
- Size limits: 95 MB per pack, 400 MB unpacked, 2 MB per preview image.
- The remote configuration in index.json is restricted: the server address must match the pattern `https://*.*.workers.dev`, and category rules may only produce paths under `assets/`.

5.5 Startup before the game (preLaunch)

The mod registers a `preLaunch` entry point. Malware likes to use this to become active before Minecraft. Here it only starts a background thread that reads `release/version.json` from GitHub and compares the version number. Nothing is downloaded, written or executed.

6. Cloud repository, packs and build environment

Component	Finding
24 release JARs (12 MC versions × 2 variants)	No foreign classes, no DLL/EXE, no embedded JARs. The builds for other MC versions contain small compatibility classes (e.g. WorldHook, compat/Click, ShieldContext). The bytecode scan shows no additional sensitive calls there and no additional URLs apart from the respective version.json.
301 texture packs	All match the size and SHA-256 in index.json exactly. Content: PNG, JSON, .mcmeta, .properties. Plus Windows leftovers (desktop.ini, Thumbs.db) and two disabled .properties files. No executable files, no path tricks.
301 preview images	Valid PNG files with no data appended after the end of the image.
tools/stylescloud.py	The author's publishing script. Only calls git (pull/add/commit/push) in the author's own repository. Not used by the mod.
tools/__pycache__/...pyc	Compiled older version of the same script (bytecode compared without execution): no functions or names that are not also in the .py.
gradle-wrapper.jar	Official Gradle wrapper; SHA-256 matches the checksums published by gradle.org for Gradle 8.7/8.8.
build.gradle / settings.gradle	Standard Fabric build without custom tasks; dependencies only from Fabric, Mojang and TerraformersMC (ModMenu).

7. Privacy notes (not malware)

The following points are not malicious functions, but you should be aware of them. They apply equally to both website variants.

Telemetry including the Minecraft name, cannot be turned off

Every 10 minutes, `StylesStats` sends the **Minecraft name** and a SHA-256 hash of a randomly generated device ID to the author's worker (`/ping`). The README describes this as "anonymous usage stats". Because the name is included, it is not anonymous. There is no switch for it in the code. The device ID is random and contains no hardware characteristics.

Presence in the tab list and lookup of other players

On a server, the mod regularly sends your own player UUID to `/presence`. It also sends the UUIDs of the players in the tab list (up to 100) to `/presence/lookup` in order to display their icons. This tells the author's server who is on a server with whom. Turning off "Show in tab" (default: on) only hides your own icon; the ping and lookup keep running.

Remote blocking

Via `blocked.json` in the cloud repository, the author can block individual accounts. The mod then no longer shows a menu and disables its own resource packs. Nothing is deleted and nothing else on the PC is changed.

Remote configuration

Via `index.json`, the author can change the server address and add new style categories without releasing a new mod version. Both are tightly restricted in the code (see 5.4); program code cannot be injected this way.

Spotify → Deezer/iTunes (Spotify variant only, off by default)

If the Spotify display is turned on, the artist and title of the current song are sent to Deezer or Apple to find a cover if Windows does not provide one.

Outdated description text

`fabric.mod.json` still says "Keeps itself up to date". However, since 1.6.0 the code no longer updates itself. The file name "auto-update" is misleading as well (see section 4).

> **What the code cannot tell us**
>
> What the author's worker does with names, UUIDs and device hashes cannot be checked from the outside. A code comment says that only the author sees the names and that no IP addresses are stored; this cannot be verified.

8. Limitations of the analysis

- **Static analysis:** The mod was not launched in Minecraft and its network traffic was not captured. However, the byte-identical build guarantees that exactly the source code that was read is what runs.
- **Server side:** The Cloudflare Worker and the website baulooprac.xyz are not part of the repositories and were not examined.
- **Future versions:** The analysis applies only to the files with the checksums in Appendix A. New versions must be reviewed again. Because the mod does not update itself, the code only changes when you install a new file yourself.

- **JARs for other MC versions** (release/mc/...) were checked with the bytecode scan but not rebuilt and not compared byte for byte; they are not relevant for MC 1.21.11.
- **Image contents** of the packs were checked for file type and appended data. Minecraft only processes them as textures; they are not executable.

9. Recommendations

- Only use files whose SHA-256 matches Appendix A. On Windows: `Get-FileHash <file>` in PowerShell.
- If you don't need the Spotify display, use the variant **without** “auto-update” in its name. It contains less code with system access. The telemetry is the same in both variants.
- If you don't want to be shown in the tab list, turn off “Show in tab” in the extras. Note that the ping and lookup keep running.
- If you want to avoid the telemetry entirely, you can block the domain `bauloo-styles-cloud-worker.workers.dev` in your firewall or hosts file. Cloud styles will keep working via GitHub; statistics, tab icons, presets and VIP features will not.
- For future updates: compare source code and JAR again, ideally with your own build.

10. Independence and disclaimer

Independence

This review is an **independent investigation out of personal interest**. Riyu(FPS) examined the mod together with Claude Code (Opus 5.5), an AI assistant by Anthropic, independently of the creator. It is based exclusively on publicly available files: the two GitHub repositories and the downloads from the official website.

- Riyu(FPS) is **not affiliated with the creator of the mod**, neither with the creator personally nor with their Discord server, website or other projects.
- There is **no collaboration, no commission and no other connection** with the creator.
- The creator was not involved in the review and had no influence on the content or result of this report.
- This report is not an advertisement for the mod and not a recommendation to use it.

Disclaimer

- **Snapshot:** The result applies exclusively to the files identified by SHA-256 in Appendix A, as of the time of review: 20 October 2026, 00:53 CEST. Other versions, other download sources or modified files are not covered.
- **No warranty for the future: No warranty** is given that no malicious code will be added in future versions or updates of the mod. The creator can change the code at any time. This also applies to the creator's servers (Cloudflare Worker, GitHub repository, website), whose behavior can likewise change at any time. Every new version must be reviewed again.
- **No guarantee:** The review was carried out to the best of our knowledge using the methodology described in this report. A static analysis cannot rule out malicious behavior with absolute certainty (see section 8).
- **No liability:** Use of the mod is at your own risk. No liability is accepted for any damage arising from the use of the mod or from reliance on this report.

11. Contact

Questions, comments or corrections regarding this report can be sent to Riyu(FPS):

Channel	Contact
Discord	@RiyuFPS
GitHub	https://github.com/ryufps
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Appendix A: Checksums (SHA-256)

Website files and Gradle wrapper:

File	SHA-256
bauloo-styles-1.6.0-mc1.21.11-auto-update.jar	20684bceb237774ae1bc9b208216b057d3f93fbebdf9b4cfa72cdb5856d6c19f
bauloo-styles-1.6.0-mc1.21.11.jar	3056d0be0cb5e4daa3d3b7830be551036a0e4c3b4976e922fe1b531cafa23b7a
gradle/wrapper/gradle-wrapper.jar	cb0da6751c2b753a16ac168bb354870ebb1e162e9083f116729cec9c781156b8

Release JARs in the cloud repository:

File (release/...)	SHA-256
bauloo-styles-no-auto-update.jar	3056d0be0cb5e4daa3d3b7830be551036a0e4c3b4976e922fe1b531cafa23b7a
bauloo-styles.jar	20684bceb237774ae1bc9b208216b057d3f93fbebdf9b4cfa72cdb5856d6c19f
mc/1.21.4/bauloo-styles-no-auto-update.jar	98799a3611816185f989dfeb5f2e856152bd61f128bbdba6ab28295fb945d90f
mc/1.21.4/bauloo-styles.jar	e50cd4aba5771e01aba40794c66e40773c8d4122189dca766411c47350c31426
mc/1.21.5/bauloo-styles-no-auto-update.jar	ca4a51a949659800bb401c04642578c471f552e798f7f7a8c6ad7201f3287a1c
mc/1.21.5/bauloo-styles.jar	352ea6774ff857b8bb8da3a6029ba6da4213a695659406fd349a1bceac1f346a
mc/1.21.6/bauloo-styles-no-auto-update.jar	8ff1ed4cc65a835f91bdb4deadadfbe38596c58b5711fbb71df1fa56783c5ad6
mc/1.21.6/bauloo-styles.jar	25687de1463f1f4c63eee9b24f62d8446177e268325ae9a9870fc5b1d11ea26f
mc/1.21.7/bauloo-styles-no-auto-update.jar	577269b2327aee2830406c5fcc2d825e1f315f2b2a71f46544b06239e726ea05
mc/1.21.7/bauloo-styles.jar	a97b9f4d9cbc302601ed255458220f24c2d4ea4bd1b3ba1dd01027325110c4c9
mc/1.21.8/bauloo-styles-no-auto-update.jar	541a1da7d4f84f4fd4bff388d77a65cc08344d971d0b868f6828d6bc6bd0e08d
mc/1.21.8/bauloo-styles.jar	d41c76e2d371739ddd45204ab8e1d044fe5d4394a2f1e5046fcd4151d5630636
mc/1.21.9/bauloo-styles-no-auto-update.jar	58f5236cd44c2aacec83d43990bfd0450f2822397a0542c48fe1f89132517608
mc/1.21.9/bauloo-styles.jar	1b960c95129934e368a53180c03e29bdb9d9ffb11c30050a66bd9c24953ccd20
mc/1.21.10/bauloo-styles-no-auto-update.jar	46ab9ea644baa1f9176182cfd0d456c2e83cf54d1d2aeada60c3ced7bcc2df65
mc/1.21.10/bauloo-styles.jar	b2b3f3689e7ae54a94349ac222b645a5c03916c8ea119e8cbfb44b86c85b0545
mc/26.1/bauloo-styles-no-auto-update.jar	5e8a8db1fab6a6da6b6c5f88435f312527b6ae98dad304cebb9932572e4bb32c
mc/26.1/bauloo-styles.jar	79c9d928c93e0a02da256a37fe79312565032fc801e4ec8202a71a88722236ba
mc/26.1.1/bauloo-styles-no-auto-update.jar	e3cd2756d9dc0cc5082c9d9cfacc2e9e918b0ac742e5b34f8b708521653ec518
mc/26.1.1/bauloo-styles.jar	177ad54b6940111db0b0b75de644e07d16db9aa99524cb5e9c711302b656cfc5
mc/26.1.2/bauloo-styles-no-auto-update.jar	65dc8b16a342305f2410ae194e0b7bd7d3cf58df21bda743c8904dab966d4c04
mc/26.1.2/bauloo-styles.jar	53c80483bf53dd22b60b1a578ade8baf04a21e78fc5704f3004eb722b5f2c371
mc/26.2/bauloo-styles-no-auto-update.jar	d92cce82f85dfab96925f75541ef1f06c77c4a6c36ba91c60cd2146fca33e206
mc/26.2/bauloo-styles.jar	3aab151031a5f7a58a40ea977063a0f28257b69f5258c8e4b7650b2ad13dd528

Appendix B: Tools and reproducibility

Tool	Version / source
Vineflower (decompiler)	1.11.1 from Maven Central, SHA-1 77767cdbc76d116c6d5e1565ec387e531085b9f4 (verified)
JDK	Eclipse Adoptium OpenJDK 21.0.12
Gradle / Fabric Loom	9.6.1 / 1.17.16 (from the repository, isolated Gradle cache)
Analysis scripts	Custom Python scripts (constant pool parser, JAR comparison, pack scanner, .pyc comparison), run with python -I; none of the examined code was loaded